AVINASH KARKADA ASHOK

Bioinformatics Analyst | Genomics Data Analysis | Reproducible Workflows

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Bioinformatics and computational biology professional with 4+ years of experience in genomics data analysis, NGS workflows, immune profiling, and reproducible computational research. Strong in R, Python, Bash, Snakemake, Git, Linux/HPC, and data visualization, with hands-on experience building analysis pipelines, QC workflows, cohort-level statistical analyses, and manuscript-facing figures for disease-focused research. Author on 12+ peer-reviewed publications across genomics, bioinformatics, structural biology, and translational disease biology.

EXPERIENCE

Bioinformatics Engineer

Johns Hopkins University School of Medicine

Sep 2024 – Present | Baltimore, USA

- Lead computational analysis for an R01-funded **PhIP-Seq** antibody-profiling project across colorectal cancer, polyp, and healthy cohorts, using R/Python-based statistical analysis, QC, visualization, and biological interpretation to support collaborator discussions and manuscript development.
- Design and maintain reproducible computational workflows for InFlux, a U01-funded highly multiplexed NGS-based influenza humoral immunity assay, including QC, validation logic, cohort-level reporting, and Linux/HPC execution in **Dr. Benjamin H. Larman's** lab.
- Build analysis-ready figures, interactive reports, and interpretation summaries that connect high-dimensional immune-profiling data to disease-focused biological questions.

Summer Bioinformatics Intern

Infinity Bio, Inc.

June 2025 – August 2025 | Baltimore, USA

- Benchmarked and optimized NGS-based **MIPSA** analysis pipelines on AWS EC2 by evaluating alignment strategies and improving I/O handling, reducing runtime from approximately 48 hours to 6 hours and cutting compute costs by approximately 87.5%.
- Built **Latch Bio**-based reproducible workflows, including a UMAP-driven sequence-space analyzer and visualization module for client-facing peptide analysis.
- Co-developed EpitopeFindeR 2, a reproducible peptide-sequence pipeline integrating FoldDisco-based 3D structural mapping to identify continuous and discontinuous epitopes from library peptide data.

Junior Research Fellow

Indian Institute of Science

Nov 2023 – Aug 2024 | Bangalore, India

- Co-first-authored a comparative genomics study decoding mammalian adaptive signatures in clade 2.3.4.4b H5N1 across non-human hosts, identifying convergent mutations relevant to host adaptation (published in *Microbiology Spectrum*).
- Contributed to the discovery and characterization of viral RHIM motifs, including a SARS-CoV-2 Nsp13 RHIM, and their role in species-specific ZBP1–RIPK3 cell-death signaling (published in *Cell iScience*).
- Integrated sequence analysis, virology, and mechanistic interpretation to support studies of influenza reassortment, host-pathogen biology, and disease-relevant viral adaptation.

Research Associate

Sciomics Pvt. Ltd. / Insilicomics Pvt. Ltd.

May 2020 – Sep 2023 Ooty, India

- Built Python programs, Bash scripts, and GUI tools that simplified recurring modeling, genome informatics, and analysis workflows in an applied research setting.
- Supported genome informatics, virtual screening, and molecular modeling projects while training students in practical bioinformatics and computational workflow basics.

EDUCATION

M.S. Bioinformatics

Johns Hopkins University

Aug 2024 – May 2026 Baltimore, USA

B.E. Biotechnology

Visvesvaraya Technological University

June 2019 – July 2023 Karnataka, India

First Class with Distinction

SKILLS

Programming, Statistics & Reproducible Workflows

R | Python | Bash | SQL | Git/GitHub | Linux/Unix | HPC/SLURM | Snakemake | Docker | Singularity | Conda/Mamba | Jupyter | R Markdown | Quarto | Shiny | AWS EC2/S3 | Statistical data analysis | Reproducible workflows

Genomics, NGS & Transcriptomics

Genomics data analysis | NGS data analysis | Single-cell RNA-seq | scRNA-seq | RNA-seq | PhIP-Seq | WGS/WES | ChIP-seq | Seurat | Scanpy | Bioconductor | edgeR | DESeq2 | limma | SAMtools | BCFtools | BEDTools | Bowtie2 | BWA | Minimap2 | STAR | HISAT2 | Kallisto | Salmon | FastQC | Cutadapt | GATK | IGV | Biopython | BLAST | MAFFT | HMMER

Data Analysis, Visualization & Biological Interpretation

Differential expression | Differential reactivity | Cohort analysis | Quality control | Pathway analysis | Enrichment analysis | Network analysis | PCA | UMAP | t-SNE | Clustering | pandas | NumPy | SciPy | scikit-learn | XGBoost | matplotlib | plotly | ggplot2 | dplyr | tidyr | GraphPad Prism | BioRender

Additional Computational Biology

Comparative genomics | Immune profiling | Host-pathogen biology | Epitope mapping | Sequence analysis | Molecular dynamics | Protein structure analysis | GROMACS | AMBER | PyMOL | UCSF ChimeraX | AlphaFold | RDKit

RELEVANT PUBLICATIONS

- **Decoding non-human mammalian adaptive signatures of clade 2.3.4.4b H5N1 to assess its human adaptive potential**
  R. Nataraj, **A. K. Ashok**, A. A. Dey, and S. Kesavardhana
  Microbiology Spectrum.

- **High-throughput screening and molecular dynamics simulations of natural products targeting LuxS/AI-2 system as a novel antibacterial strategy for antibiotic resistance in *Helicobacter pylori***
  A. K. Ashok, T. S. Gnanasekaran, H. S. S. Kumar, K. Srikanth, N. Prakash, and P. Gollapalli
  Journal of Biomolecular Structure and Dynamics.

- **Bat RNA viruses employ viral RHIMs orchestrating species-specific cell death programs linked to Z-RNA sensing and ZBP1–RIPK3 signaling**
  S. Mishra, D. Jain, A. A. Dey, S. Nagaraja, M. Srivastava, O. Khatun, K. Balamurugan, M. Anand, **A. K. Ashok**, S. Tripathi, *et al.*
  Cell iScience.

- **Computational-driven discovery of AI-2 quorum sensing inhibitor targeting the $5'$-methylthioadenosine/S-adenosylhomocysteine nucleosidase (MTAN) to combat drug-resistant *Helicobacter pylori***
  M. Kumar, **A. K. Ashok**, T. Bhat, K. Ballamoole, and P. Gollapalli
  Computers in Biology and Medicine.

ACHIEVEMENTS & SERVICE

- 12+ peer-reviewed publications as first- or co-author across influenza genomics, viral cell-death signaling, computational biology, and computational drug discovery.
- Owned collaborator-facing analysis outputs, figures, and manuscript-facing interpretation for immune-profiling projects at Johns Hopkins University School of Medicine.
- Paper presentation at Immunology 2025 (AAI Annual Meeting).
- Peer reviewer for *Nature Communications*, *Journal of Crohn's and Colitis*, *J. Biomolecular Structure and Dynamics*, and *J. Biological Macromolecules*.
- Awarded Best Paper Presentation (JAIVIK 2022) and Best Undergraduate Thesis.
- Mentored multiple students in practical bioinformatics, molecular dynamics, and NGS analysis.